

Home Lab ITF-024

Lab Objectives

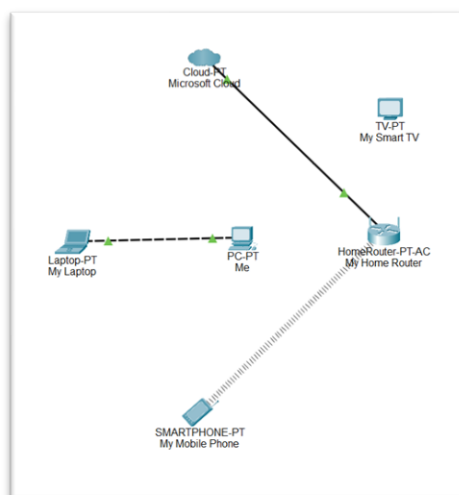
Develop a practical understanding of what networks are, why they exist, and how devices communicate within modern environments using Cisco Packet Tracer to build, expand, and organise simple networks, observe how devices connect, and relate network topologies to real-world environments. This will develop foundational networking concepts that support the remainder of the networking pathway.

Section 1 – What Networks Do I Use?

Network	Devices Connected	Services/Apps Used
Home Wi-Fi	Laptop, phone, smart TV	Netflix, browser, email
Mobile network (4G/5G)	Phone	Maps, messaging, calls
Work/School network	Laptop	Email, intranet, shared drives
Public Wi-Fi	Phone, Laptop	Casual browsing only

Reflection:

I usually interact with around four different networks daily. My home network belongs to me, but my mobile network, work network, and public Wi-Fi all belong to someone else. I trust my home and work networks the most because they require sign-in and I know who manages them. On the other hand, Public Wi-Fi is the one I don't trust. The diagram below is a diagram of my home network using Cisco Packet Tracer.



Home Network Path

Phone ➡ Home ➡ Wi-Fi ➡ Home Router ➡ ISP Internet Streaming Service

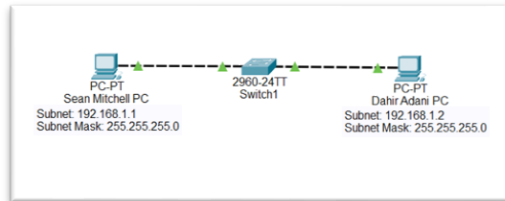
Work Network Path

Laptop ➡ Work Network ➡ Work Router/Firewall ➡ Internet ➡ Email Server

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Section 2 - Building Your First Network

Setup: Two PCs (PC0, PC1) connected to a single switch, IPs 192.168.1.1 and 192.168.1.2, subnet 255.255.255.0. Ping from PC0 to PC1 succeeded with replies.



```
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=11ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=4ms TTL=128
Reply from 192.168.1.1: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 11ms, Average = 6ms

C:\>Ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Observations:

1. Devices are represented as small icons with labelled ports; the switch shows multiple numbered ports (FastEthernet0/1, 0/2, etc.).
2. Connections appear as straight lines between specific ports, and the link ends turn green once both sides are configured with a valid interface state.
3. The devices are organized in a simple star pattern, with the switch in the middle and each PC on either side.

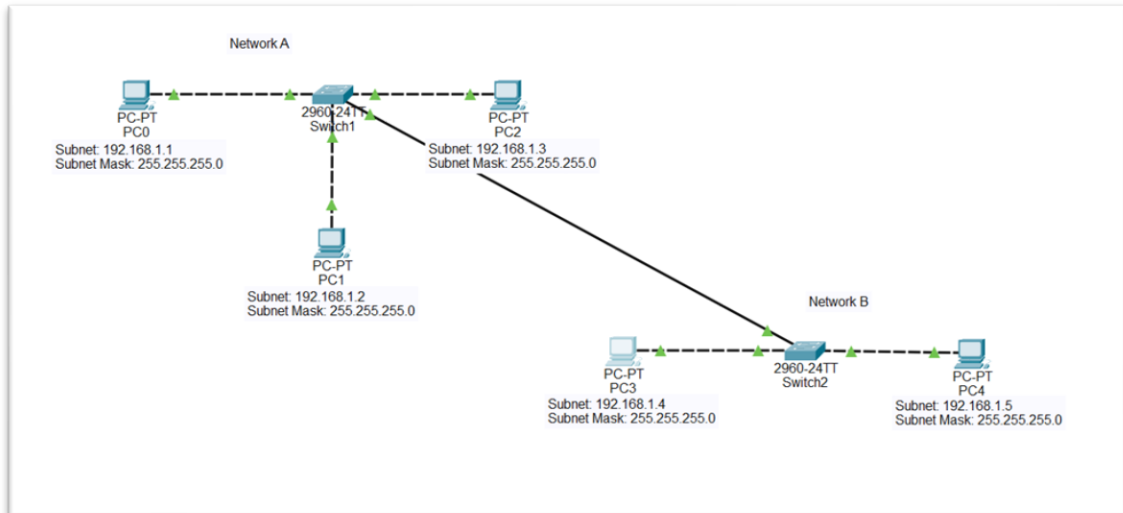
Why the switch exists: It gives each PC a shared point to connect through instead of requiring a dedicated cable between every pair of devices. Even with only two PCs it works, but its purpose becomes clearer once more devices are added.

If the switch was removed: The two PCs would have no way to communicate unless directly cabled to each other, since neither has a free port to link to the other.

Why go through the switch instead of directly connecting PCs: A direct cable only supports two devices. A switch allows the network to scale - more devices can be added later without changing the existing wiring.

Section 3 - Growing a Network

Setup: Expanded to 5 PCs and 2 switches, with the two switches connected to each other. All PCs given IPs in the same 192.168.1.0/24 subnet.



Observations: As more devices were added, the diagram became noticeably harder to read. Cables started to overlap, and it took more effort to figure out which PC connected to which switch. The network didn't become functionally more complicated to use (ping still just works), but it became visually harder to manage.

Reflection: A home network usually stays small and rarely needs a diagram, but a larger organizational network with dozens or hundreds of devices absolutely needs one otherwise, troubleshooting a fault or planning a change becomes guesswork. Diagrams become more important as environments grow because no single person can hold the entire structure in their head.

Section 4 - Connecting Networks Together

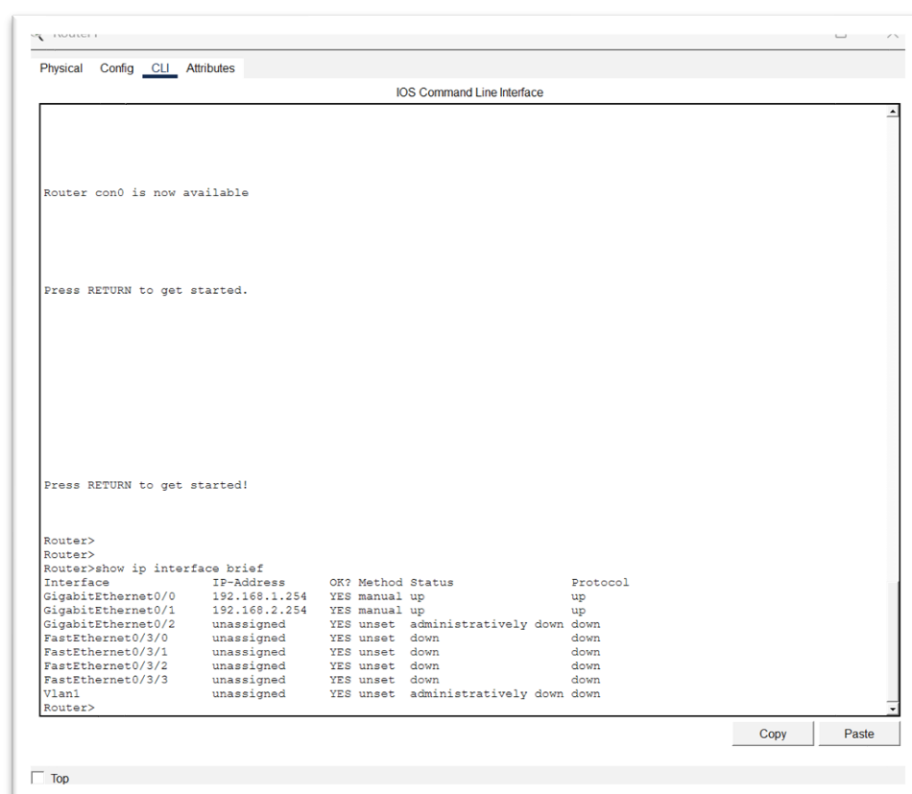
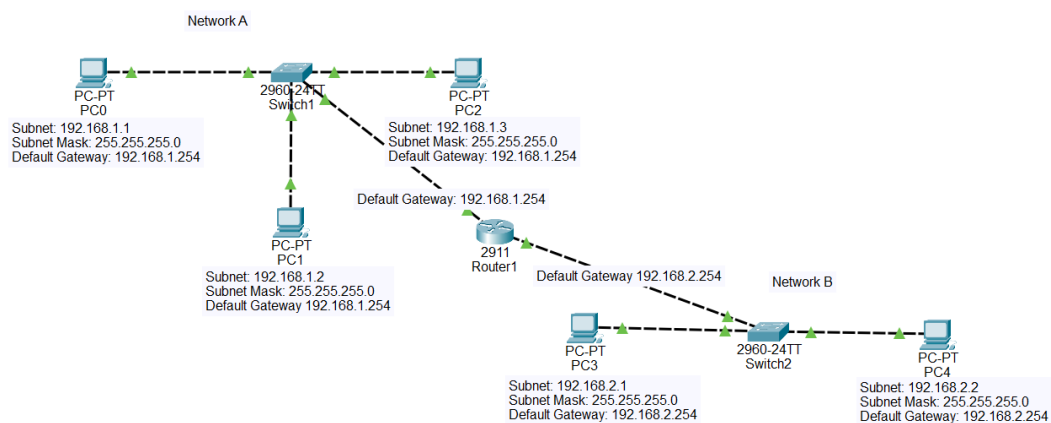
Setup: Added a second switch group (Network B: 192.168.2.0/24) and connected it to the original network (Network A: 192.168.1.0/24) via a router. Router interfaces configured as gateways for each side (192.168.1.254 and 192.168.2.254). Ping from a PC in Network A to a PC in Network B succeeded after setting default gateways.

Observations:

1. The two networks appear as two distinct clusters on the canvas, with the router positioned between them.
2. The router has a different icon from the switches and shows two separate interfaces, one facing each network.
3. Adding the router noticeably changed the topology from "one flat network" to "two networks joined together."

Why organizations use multiple networks: Separating networks helps with organization, security, and traffic management, for example, keeping finance devices separate from guest Wi-Fi.

Why a router is needed: A switch alone can't connect two different subnets; a router is what makes forwarding traffic between separate networks possible.



Section 5 - Relating Networks to the Real World

Environment identified: Small business office, the size (roughly 8-10 devices across two subnets with a router) is too large for a typical home network but far smaller than an enterprise. It likely represents a small office with a few departments and shared internet access.

Updates made: Renamed devices to reflect their role (Reception PC1, Sales PC2, Finance PC3, HR PC4, Security Office PC5, Office 1 Switch and Office 2 Switch), repositioned staff devices together and guest devices separately, and added a cloud device labelled "Internet" connected to the main router's third interface.

Reflection: Networks grow over time because organizations add staff, devices, and services faster than they remove them. Diagrams stay useful because they let anyone (not just the person who built the network) understand and troubleshoot it. Different organizations design networks differently based on their size, security needs, and budget; a school's network priorities (content filtering, many simultaneous users) differ a lot from a small business's (fewer users, but more sensitive financial data).

Reflection answers:

- Most important device:** The router which is the only device that connects the two internal networks to each other and to the internet. Every switch's traffic ultimately depends on it for anything outside its local group.
- What became more difficult as the network grew:** Keeping IP addressing consistent and avoiding overlapping cables in the diagram. Troubleshooting also got harder, a failed ping could now be caused by several different things (wrong IP, wrong gateway, disabled interface, wrong cable).
- If a key device failed:** If the router failed, each individual network would still work internally (PCs within the same switch group could still reach each other), but nothing could cross between networks or reach the internet. If a switch failed, every device connected to it would lose all connectivity, even to devices on the same switch.
- Scaling to twice as many devices:** I'd add another switch rather than overloading existing ports, plan out IP address ranges in advance instead of assigning them ad hoc, and consider whether a second router or VLANs (a topic for a future lab) would help keep departments logically separated as the network grows.

